CPE 314 — Software Design & Software Design Laboratory

Week 8 — Project Proposal

Group Name / No.:	Group 1 | BSCPE 3B | EDP 69323
Members:	John Benedict Alaiza
	Acexion B. Mahigugmaon
	Glyde V. Arreglado
	Tristan Dominic Urot
	Joshua Brian S. Enovero
	Eman De Paz
Repository URL:	https://github.com/XionAce/cpe314-69323-capstone-team-01
Theme:	Campus Marketplace with Reviews
API You Will Use:	FakeStoreAPI — https://fakestoreapi.com (no key required)

1. One-Paragraph Pitch

We propose UC-B Campus Marketplace with Reviews, a Java desktop application where students, faculty, school units, and department committees can post and browse goods. It retains textbook, electronics, and furniture listings with type-specific condition rules, recursive category search, buyer-interest queues, browsing history, and reviews. SQLite/JDBC persistence, a responsive background sales-history JSON import, FakeStoreAPI suggestions, and category price/rating bar charts integrate the required lessons. Uniforms, department shirts, and intramural merchandise are a proposed apparel extension for instructor approval; online payments are outside scope.

2. Entity-Relationship Sketch

PK = primary key; FK = foreign key. The sketch shows the core table relationships.

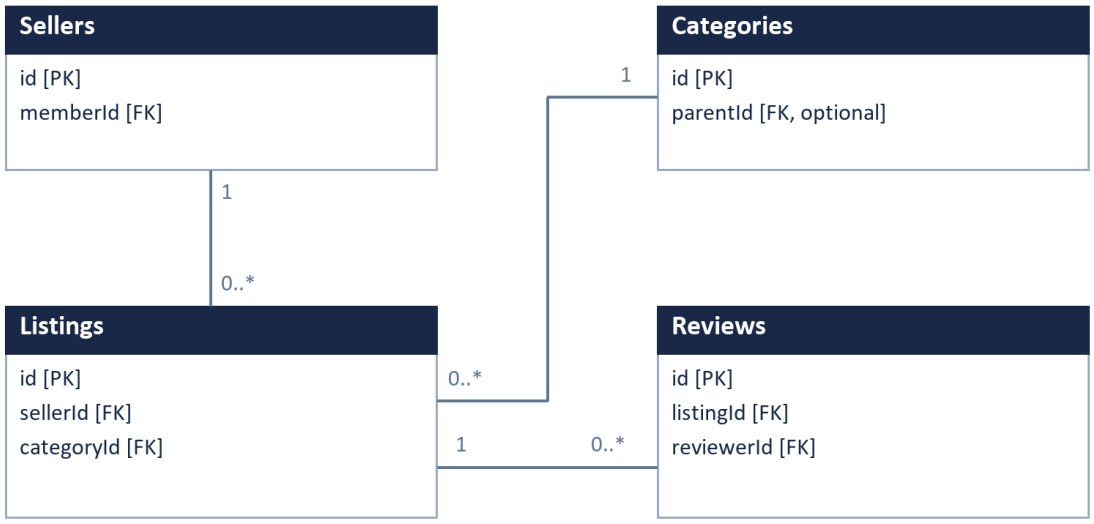


Figure 1: Core preliminary entity relationships

Each seller and category may have many listings; each listing may have many reviews. The complete preliminary data model is recorded in the project design document.

3. Feature Checklist

Check off which mandatory requirement each planned feature will satisfy:

Planned Feature	Requirement(s) It Satisfies
Abstract Listing; Textbook, Electronics and Furniture grading overrides.	Object-Oriented Design (Week 4)
FIFO interest Queue, LIFO history Stack behavior; recursive category-tree search.	Data Structures and Recursion (Week 3)
At least 6 JUnit 5 tests with normal and edge cases.	Automated Testing (Week 4)
Related SQLite tables; JDBC PreparedStatement and foreign keys.	Data Modeling and Database (Week 5)
Parse catalog products for editable category and price suggestions.	API Consumption (Week 5)
JavaFX post/search/review/import controls; background JSON/API work.	Event-Driven GUI and Concurrency (Week 6)
Mean category prices and review ratings; labeled bar charts.	Data Mining and Visualization (Week 6)

4. Task Split

Member	Primary responsibility
John Benedict Alaiza - Backend Developer (Listing Models)	Sample listings, categories and condition examples.
Acexion B. Mahigugmaon - Project Leader / Integration	Architecture, implementation planning and integration.
Glyde V. Arreglado - Database Admin	ER sketch, attributes and relationship review.
Tristan Dominic Urot - UI/GUI Developer	GUI sketches, control labels and user steps.
Joshua Brian S. Enovero - Backend & API Developer	Sample JSON and API/import validation cases.
Eman De Paz - QA & Testing	Test cases and expected analytics calculations.

Proposed roles for member confirmation; actual contributions will be recorded in each member's own commits.

5. Public API You Plan to Use

FakeStoreAPI — GET https://fakestoreapi.com/products (no key): provides product titles, descriptions, categories and prices for editable category and reference-price suggestions. Docs: https://fakestoreapi.com/docs.

6. Risks / Open Questions

How will recursive searches include listings from every nested subcategory?

What should the application show when FakeStoreAPI has no comparable product?

How will we calculate category ratings and average prices accurately?